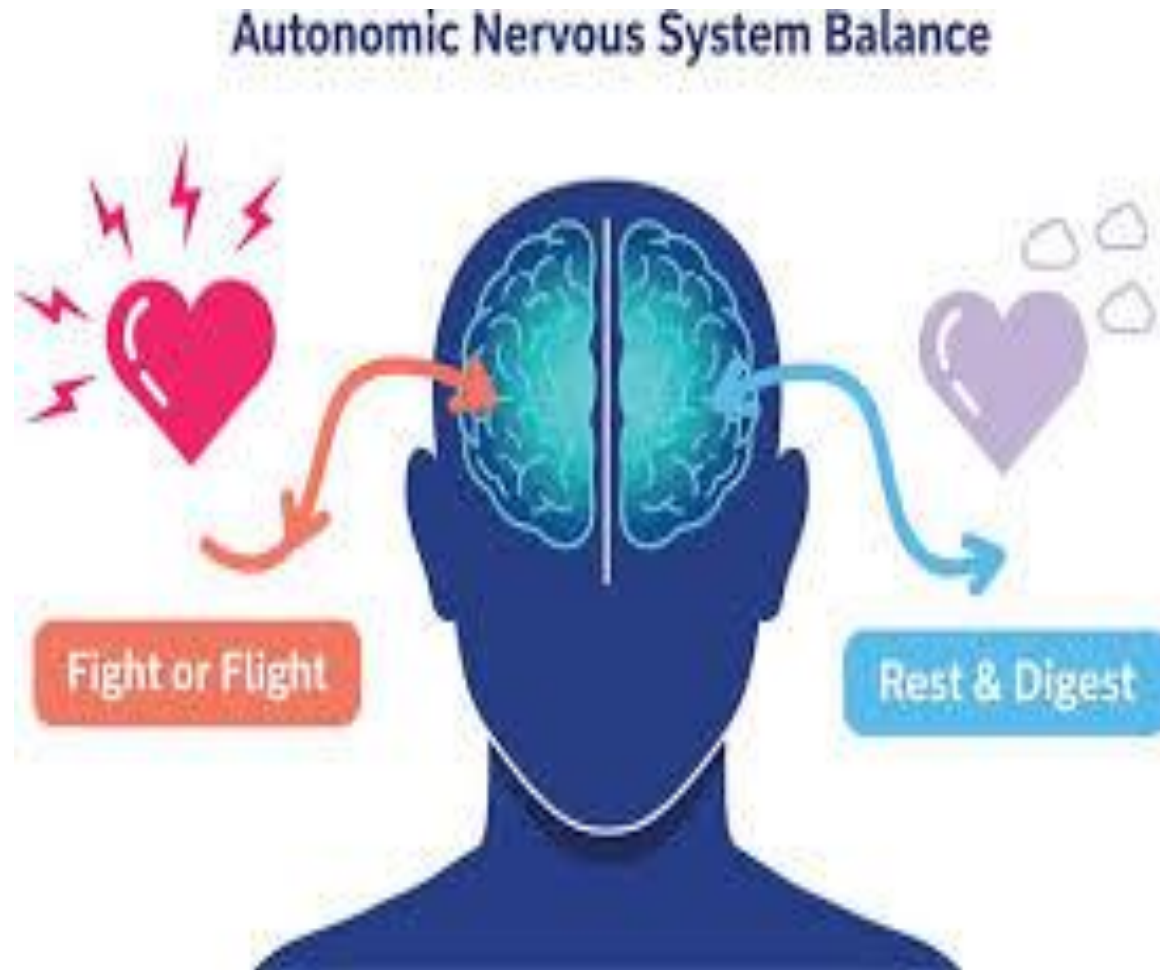


AUTONOMIC NERVOUS SYSTEM

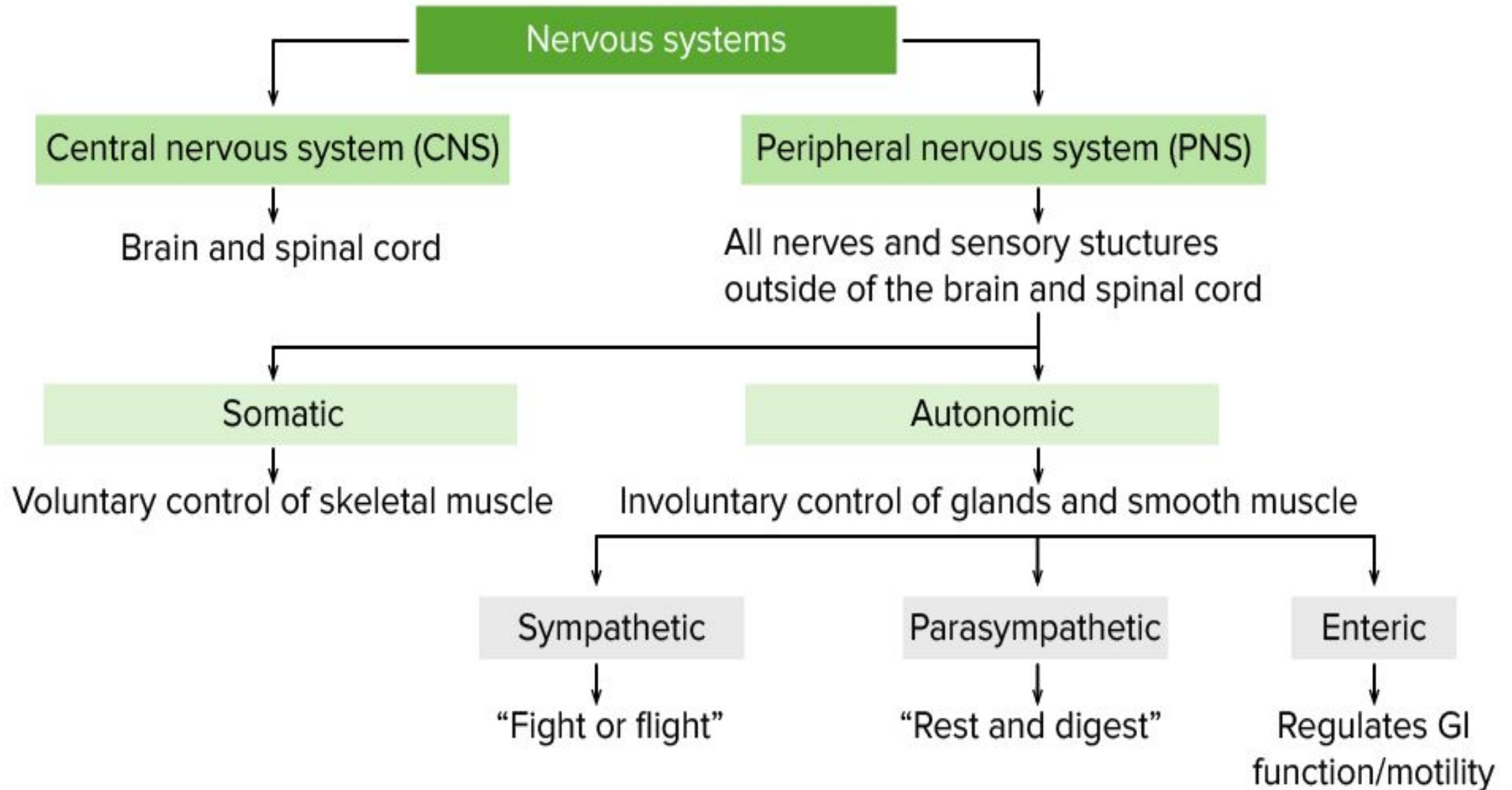


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LEARNING OBJECTIVES

By the end of this lecture students should be able to:

- Define the autonomic nervous system.
- Identify the divisions of autonomic nervous system & differentiate between sympathetic nervous system & parasympathetic nervous system.
- Compare the sympathetic & parasympathetic effects of autonomic nervous system.
- Describe the anatomical organization of autonomic nervous system.
- Identify the peripheral components i.e preganglionic & postganglionic neurons.
- Define the autonomic ganglia.
- Understand the key neurotransmitters of ANS (acetylcholine & norepinephrine).
- Understand adrenergic & cholinergic receptors.



What is autonomic nervous system?

Autonomic nervous system is the portion of nervous system that controls most visceral functions of the body.

Autonomic nervous system is a critical component of peripheral nervous system responsible for regulating involuntary physiological processes like cardiovascular function, respiration, digestion & glandular activity. Autonomic nervous system is concerned with the regulation of visceral or vegetative functions so it is also called ***vegetative*** or ***involuntary nervous system***.

This system controls involuntary functions and maintains homeostasis and regulates heart rate, blood pressure, gastrointestinal motility, gastrointestinal secretion, respiration, body temperature, sweating and many other activities.

Rapidity of autonomic nervous system

One of the most striking characteristics of autonomic nervous system is the **rapidity** and **intensity** with which it can change visceral functions.

For example, within 3 to 5 seconds it can increase the heart rate to twice normal & within 10 to 15 seconds the arterial pressure can be doubled. On the other extreme, the arterial pressure can be decreased low enough to cause fainting within 10 to 15 seconds. Similarly, sweating can begin within seconds & the urinary bladder may empty involuntarily also within seconds.

GENERAL ORGANIZATION OF ANS

The autonomic nervous system is activated mainly by centers located in spinal cord, brain stem & hypothalamus. Also, portions of the cerebral cortex especially of limbic cortex can transmit signals to the lower centers & can influence autonomic control.

CONTROL OF ANS THROUGH VISCERAL REFLEX

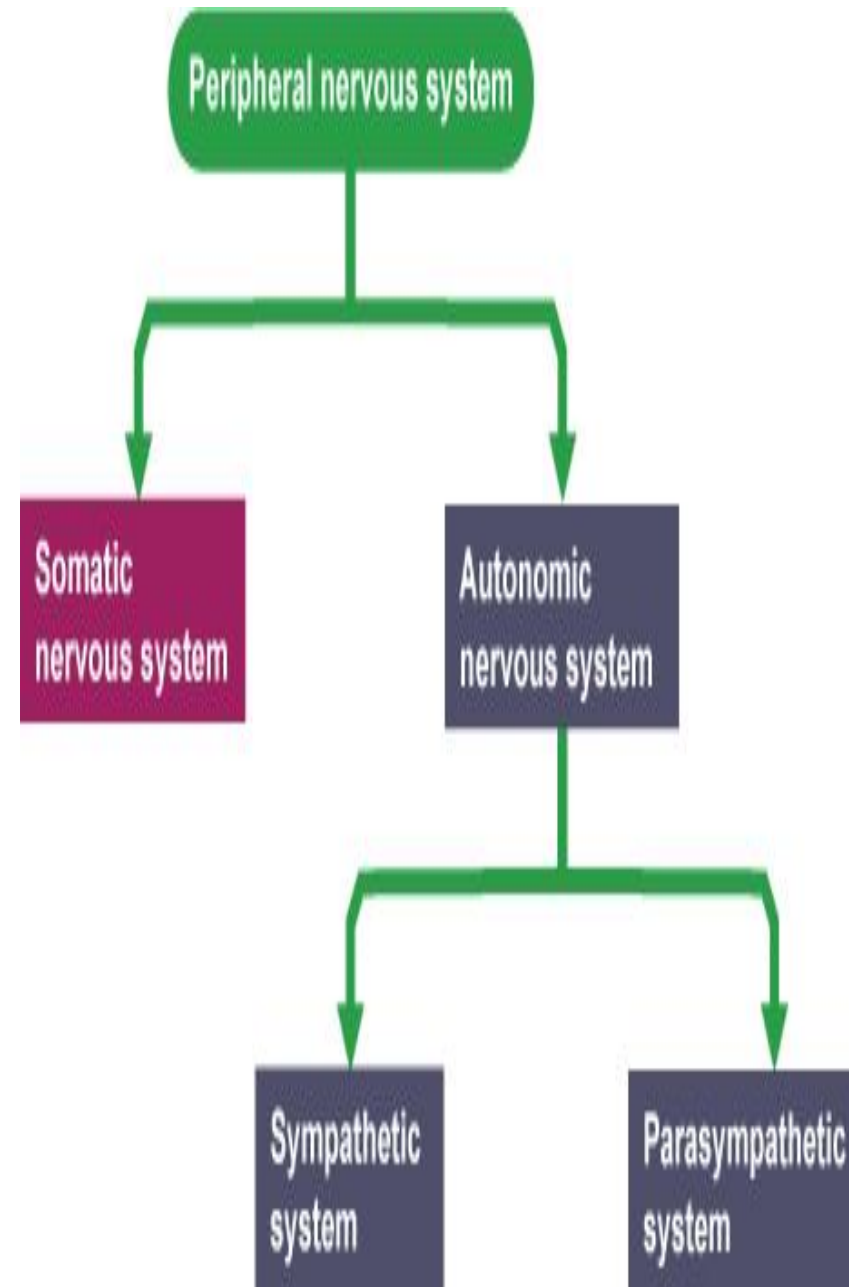
Autonomic nervous system also operates through visceral reflexes. Subconscious sensory signals from visceral organs enter the autonomic ganglia, brain stem or hypothalamus & return subconscious reflex responses back to the visceral organs & hence control their activities.

DIVISIONS OF ANS

The efferent autonomic signals are transmitted to the various organs of the body through two main divisions:

1. Sympathetic nervous system
2. Parasympathetic nervous systems

The term sympathetic & parasympathetic are strictly anatomic terms & refers to the anatomic origin of the pre-ganglionic neurons in the central nervous system.



SYMPATHETIC NERVOUS SYSTEM

Sympathetic nervous system also called **fight or flight system** is activated during stress, fear, exercise. The overall function of sympathetic nervous system is to **mobilize the body for activity**. Sympathetic nervous system is activated with a response called fight or flight which includes increased arterial pressure, increase blood flow to active muscles, increased metabolic rate, increased blood glucose concentration and increased mental activity & alertness.

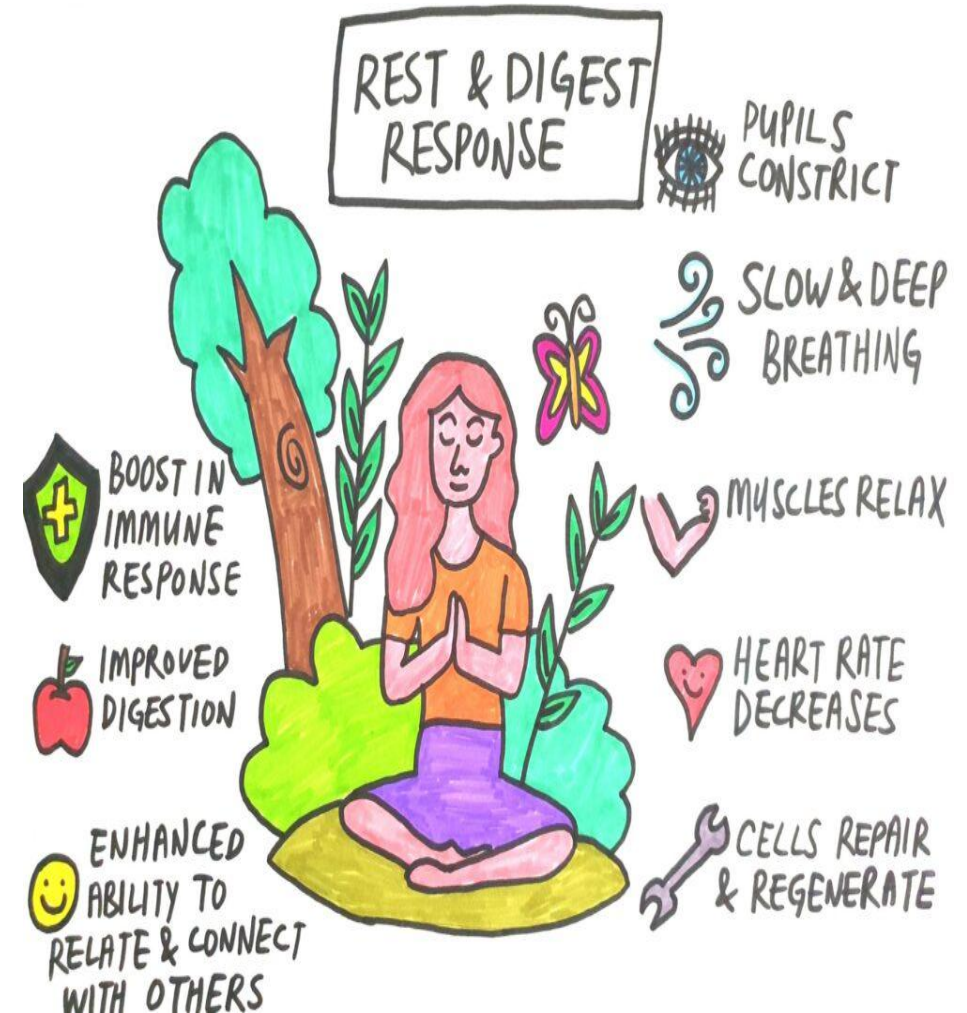


FUNCTIONS OF SYMPATHETIC NERVOUS SYSTEM

- Prepare body for stress or emergencies.
- Increase heart rate for more blood to muscles.
- Dilates pupil for better vision.
- Bronchodilation for more oxygen intake.
- Decrease digestion (redirection of energy).

PARASYMPATHETIC NERVOUS SYSTEM

Parasympathetic nervous system also known as **rest & digest** system dominates during sleep, digestion & relaxation. The overall function of parasympathetic nervous system is **restorative to conserve energy**.

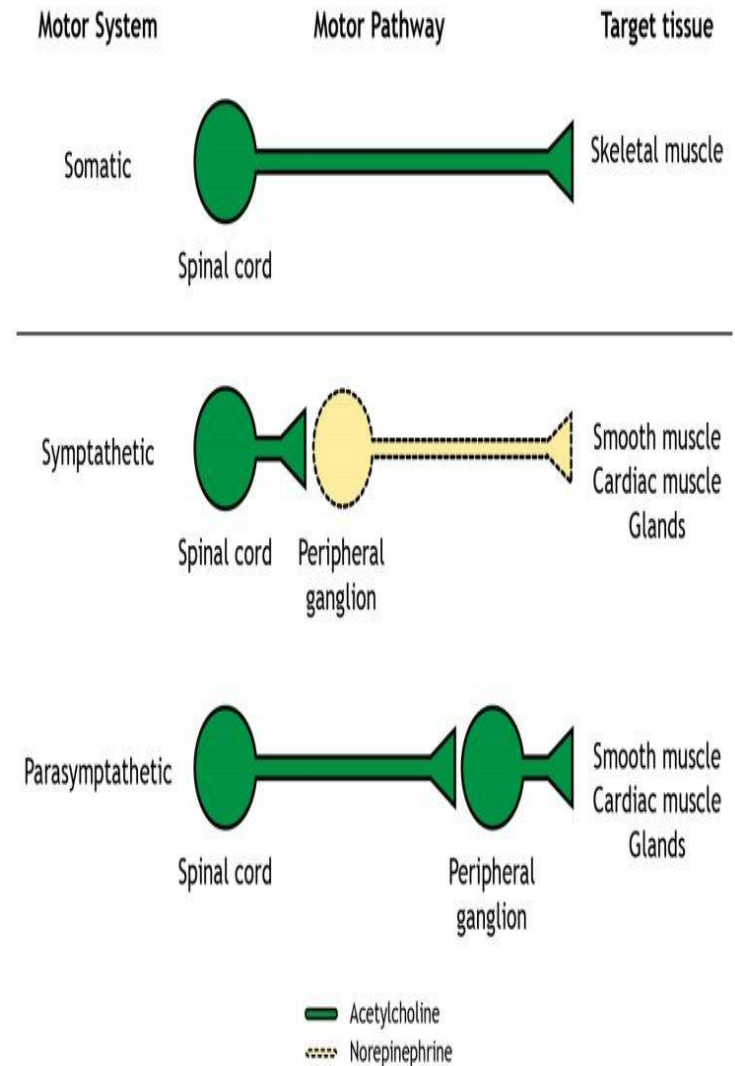


FUNCTIONS OF PARASYMPATHETIC NERVOUS SYSTEM

- Decrease heart rate(relaxation).
- Constricts pupil(normal light conditions).
- Increase digestion for more nutrient absorption.
- Promotes energy storage.

PATHWAYS IN AUTONOMIC NERVOUS SYSTEM

Unlike the somatic nervous system which uses only one neuron, autonomic nervous system uses two neuron chains (preganglionic neuron & postganglionic neuron) with a synapse in ganglion to transmit the efferent autonomic signals.



PREGANGLIONIC NEURONS **VERSUS** **POSTGANGLIONIC NEURONS**

PREGANGLIONIC NEURONS

The neurons of the autonomic nervous system whose cell bodies lay in the central nervous system and axons terminate in a peripheral ganglion, synapsing with postganglionic neurons

Arise from the central nervous system

Supply to the autonomic ganglia

Cell bodies are located in the brain or spinal cord

Axons are myelinated, type B fibers

Synapses with one postganglionic neuron

All are cholinergic

POSTGANGLIONIC NERUONS

The neurons of the autonomic nervous system whose cell bodies lay in an autonomic ganglion and whose axons terminate in a visceral effector (smooth or cardiac muscle or glands)

Arise from the autonomic ganglia

Supply to the effector organ

Cell bodies are located in the ganglion

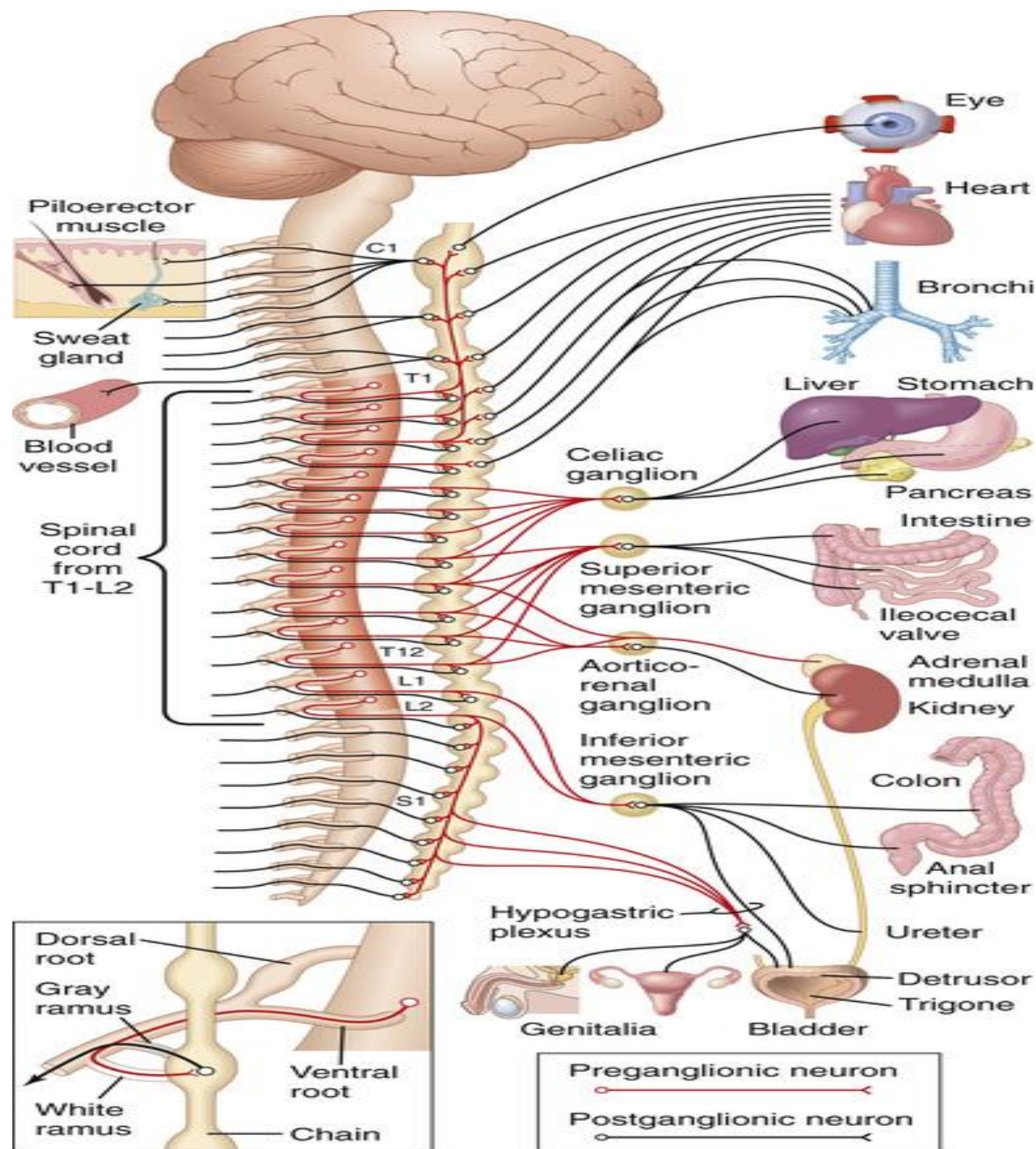
Axons are unmyelinated, type C fibers

Can synapse with more than one presynaptic neurons

Neurons in the sympathetic division are adrenergic and neurons in the parasympathetic division are cholinergic

ORIGIN OF PREGANGLIONIC NEURONS IN SYMPATHETIC NERVOUS SYSTEM

- The pre-ganglionic neurons in the sympathetic nervous system originate in the thoracic & lumbar spinal cord & leave the spinal cord via ventral motor roots & white rami & project either to the paravertebral ganglia or to a series of prevertebral ganglia. As the preganglionic neurons in the sympathetic nervous system originate in the thoracic & lumbar region of spinal cord, the sympathetic division is also called **thoracolumbar division**.
- One category of preganglionic neuron synapse on post ganglionic neuron within the sympathetic chain either at the same segmental level or the preganglionic fibres may turn in cranial or caudal direction & synapse in the ganglia at higher or lower levels in the sympathetic chain. The other category passes through the sympathetic chain without synapsing and synapse in the prevertebral ganglia outside the sympathetic chain & supply the visceral organs, glands & the enteric nervous system of GI tract.



LOCATION OF AUTONOMIC GANGLIA IN SYMPATHETIC NERVOUS SYSTEM

The ganglia of sympathetic nervous system are located near the spinal cord either in the **paravertebral ganglia** (known as sympathetic chain) or in the **prevertebral ganglia**.

The **superior cervical ganglion** projects to organs in the head such as eyes and the salivary glands. The **celiac ganglion** projects to the stomach & the small intestine. The **superior mesenteric ganglion** projects to the small and large intestine & the **inferior mesenteric ganglion** projects to the lower large intestine, anus, bladder & genitalia.

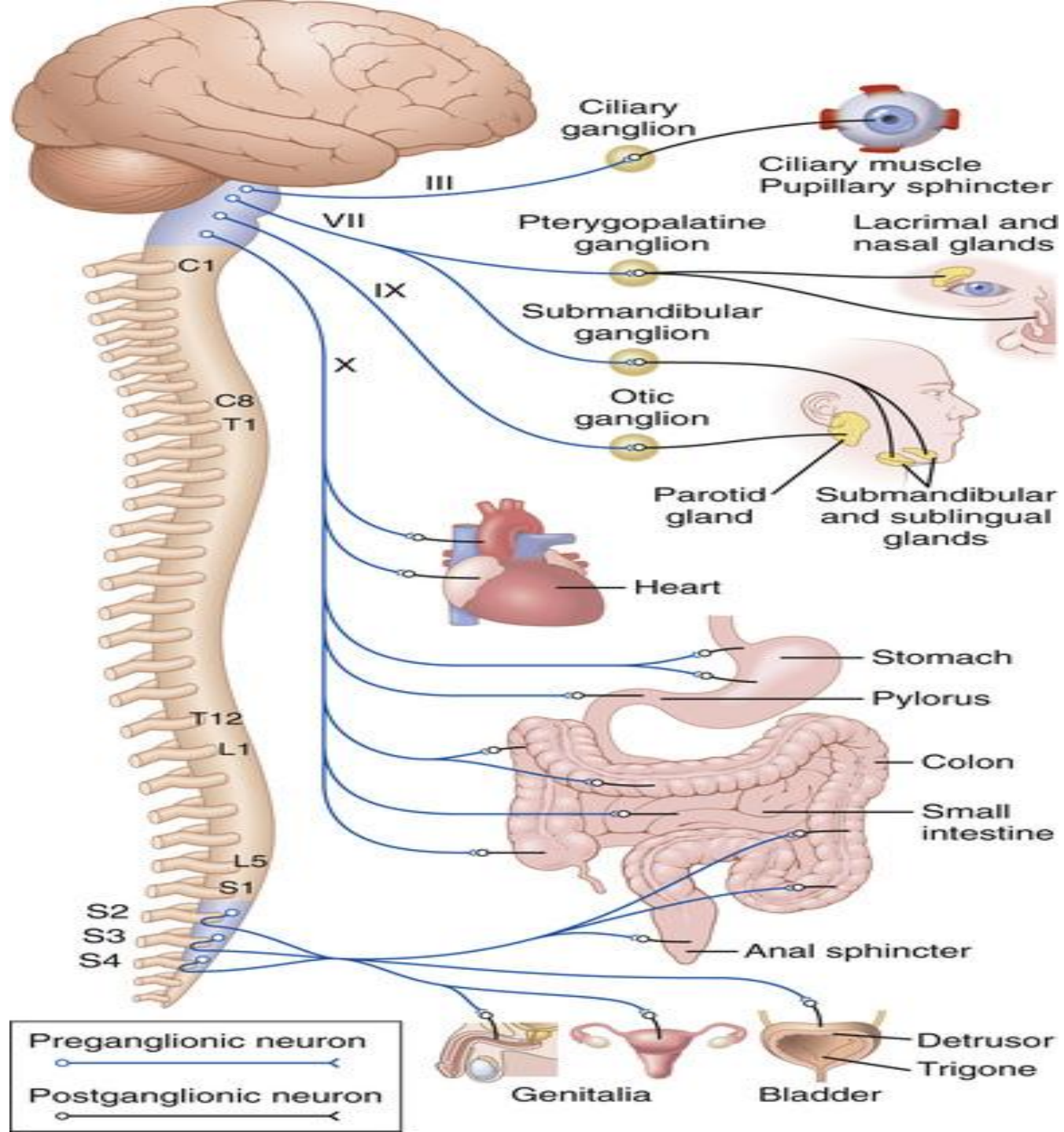
LENGTH OF PREGANGLIONIC & POSTGANGLIONIC AXONS IN SYMPATHETIC NERVOUS SYSTEM

As the sympathetic ganglia are located near the spinal cord, the preganglionic nerve axons are short & the postganglionic nerve axons are long so that they can reach the peripheral effector organs.

ORIGIN OF NEURONS IN PARASYMPATHETIC NERVOUS SYSTEM

Preganglionic neurons of the parasympathetic division arise from the nuclei of the cranial nerves(CN) III, VII, IX & X or from sacral spinal cord segments S2-S4. Therefore, the parasympathetic division is called *craniosacral division*.

As in the sympathetic division, the origin of the preganglionic neurons in the CNS is consistent with the projection to the effector organs in the periphery. The parasympathetic innervation of eye muscles originates in the Edinger-Westphal nucleus in the midbrain & travels to the periphery in CN-III. The parasympathetic innervation of the heart, bronchioles & the gastrointestinal tract originates in the nuclei of the medulla & travels to the periphery in CN-X & the parasympathetic innervation of the genitourinary organs originates in the sacral spinal cord & travels to the periphery in the pelvic nerves.



LOCATION OF AUTONOMIC GANGLIA IN PARASYMPATHETIC NERVOUS SYSTEM

In contrast to the sympathetic ganglia, which are located near the CNS, the ganglia of the parasympathetic nervous system are located near, on or in the effector organs(e.g. ciliary pterygopalatine, submandibular, otic).

LENGTH OF PREGANGLIONIC & POSTGANGLIONIC AXONS IN PARASYMPATHETIC NERVOUS SYSTEM

The relative length of the preganglionic & postganglionic axons in the parasympathetic division is the reverse of the relative lengths in the sympathetic division. This difference shows the location of the parasympathetic ganglia. As the parasympathetic ganglia are located near, on or in the effector organs, therefore the preganglionic neurons of parasympathetic division have long axons & the postganglionic neurons have short axons.

NEUROTRANSMITTERS IN AUTONOMIC NERVOUS SYSTEM

The sympathetic and parasympathetic nerve fibres secrete mainly one of the two types of the synaptic transmitter substances namely:

1. Acetylcholine
2. Norepinephrine

The fibres that secrete acetylcholine are called **cholinergic** while those that secrete norepinephrine are called **adrenergic** (derived from adrenalin which is alternate name for epinephrine).

1. CHOLINERGIC NEURONS

All preganglionic neurons are cholinergic in both the sympathetic & parasympathetic nervous system. These fibres secrete acetyl choline or acetylcholine like substances which when applied to the ganglia will excite both the sympathetic & parasympathetic postganglionic neurons.

CONT..

In addition to all the preganglionic neurons that are cholinergic, *either all or almost all of the postganglionic neurons of the parasympathetic nervous system are also cholinergic.*

2. ADRENERGIC NEURONS

In contrast to the postganglionic neurons of the parasympathetic nervous system, *most of the postganglionic sympathetic neurons are adrenergic.* However, the postganglionic sympathetic nerve fibres to the sweat glands & to a very few blood vessels are cholinergic.

SECRETIONS AT TERMINAL NERVE ENDINGS OF POSTGANGLIONIC NEURONS

- The terminal nerve endings of the parasympathetic system all or virtually all secrete acetylcholine.
- Almost all of the sympathetic nerve endings secrete norepinephrine but a few secrete acetylcholine.
- These neurotransmitters in turn act on the different organs to cause respective sympathetic or parasympathetic effects.
- Therefore, *acetylcholine* is called a *parasympathetic neurotransmitter* & *norepinephrine* is called a *sympathetic transmitter*.

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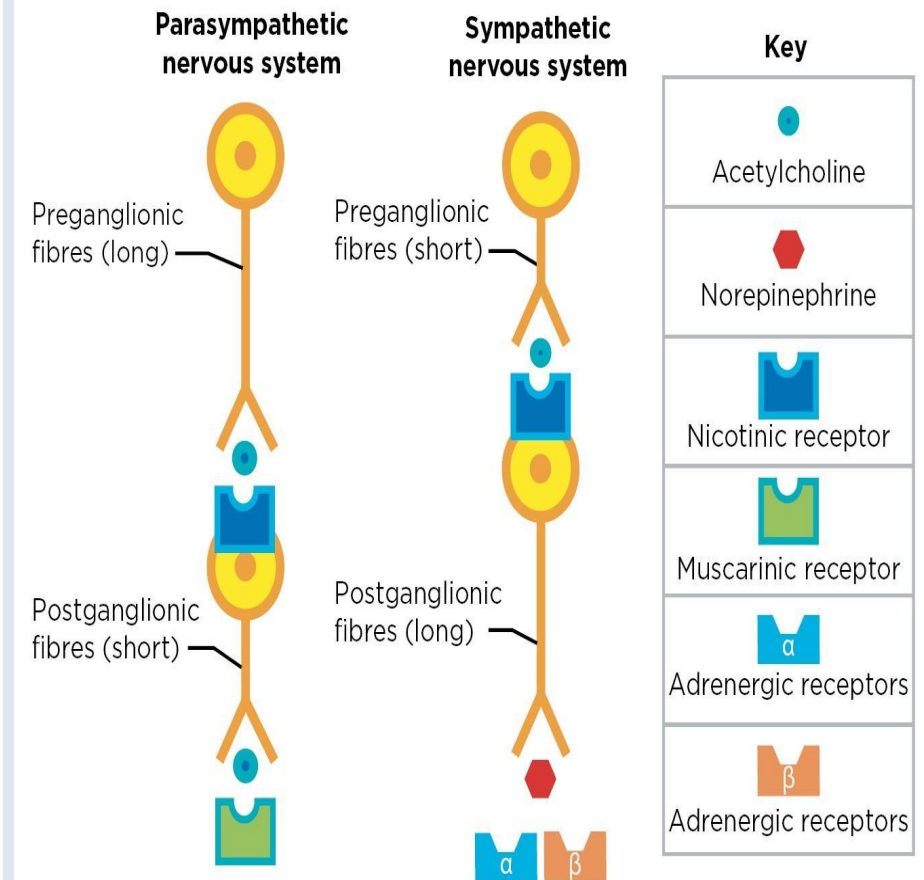
DIVISIONS		Central nervous system	Peripheral nervous system	Effector organs
Somatic nervous system				Skeletal muscle
Autonomic nervous system	Sympathetic nervous system		 Ganglion	Sweat glands
			 Ganglion	Blood vessels Glands Heart
	Parasympathetic nervous system		 Ganglion	Lungs Stomach Kidneys Bladder Intestines

TYPES OF RECEPTORS IN AUTONOMIC NERVOUS SYSTEM

Acetylcholine & norepinephrine activates one of the two principle types of receptors:

1. Cholinergic receptors
2. Adrenergic receptors

Fig 3. Neurotransmitters and receptors of the preganglionic and postganglionic neurons



TYPES OF CHOLINERGIC RECEPTORS

Acetylcholine activates mainly two types of receptors called muscarinic & nicotinic receptors.

1. Muscarinic receptors

Muscarinic receptors use G protein as signaling mechanism. Muscarinic receptors are present on all effector cells that are stimulated by the postganglionic cholinergic neurons of either the parasympathetic or the sympathetic nervous system.

2. Nicotinic receptors

Nicotinic receptors are ligand-gated ion channels found in autonomic ganglia at the synapses between the preganglionic & postganglionic neurons of both the sympathetic & parasympathetic nervous system.

TYPES OF ADRENERGIC RECEPTORS

There are two types of adrenergic receptors:

1. Alpha receptors(α_1 , α_2)
2. Beta receptors(β_1 , β_2 , β_3)

ALPHA RECEPTORS VERSUS BETA RECEPTORS

Alpha receptors are the cell receptors that control physiological processes like vasoconstriction, intestinal relaxation, pupil dilation upon interaction with epinephrine and norepinephrine	Beta receptors are a group of receptors that control vasodilation, relaxation of the bronchial and uterine smooth muscles, and increased heart rate
Stimulate effector cells	Relax effector cells
The two types are alpha 1 and alpha 2	The 3 types are beta 1, beta 2, and beta 3
Mainly occur in vascular smooth muscles and effector tissues	Mainly occur in bronchi muscles, heart muscles, and uterine muscles
Stimulate smooth muscles	Stimulate both cardiac and smooth muscles
Methoxamine is an example of alpha 1 agonist while clonidine is an example of alpha 2 agonists	Albuterol is an example of beta 1 agonist while metoprolol is an example of beta 2 agonists
	Visit www.pediaa.com

ADRENERGIC RECEPTORS VERSUS CHOLINERGIC RECEPTORS

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ADRENERGIC RECEPTORS

Receptors on the surface of cells that get activated when they bind a type of neurotransmitter called a catecholamine

Sympathetic nervous system stimulates adrenergic receptors

Bind to catecholamines: epinephrine, and norepinephrine

G-protein coupled receptors

Glycoproteins

Types: Alpha and beta receptors

Stimulate the 'fight or flight' response

CHOLINERGIC RECEPTORS

Receptors on the surface of cells that get activated when they bind a type of neurotransmitter called acetylcholine

Parasympathetic nervous system stimulates cholinergic receptors

Bind to acetylcholine

Inotropic-ligand-gated and metabotropic receptors

Integral membrane proteins

Types: Nicotinic and muscarinic receptors

Stimulate the 'digest and rest' response

Parasympathetic

Pupil
Constriction

Heart
Slow heartbeat

Airways
Constriction

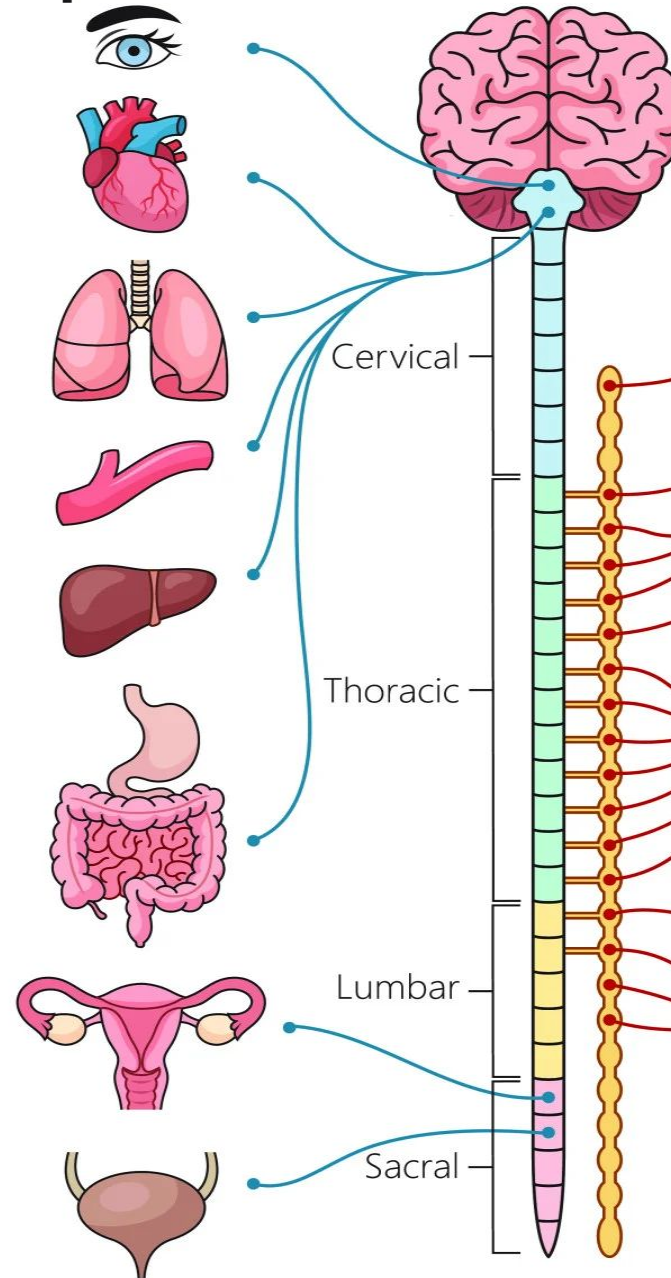
Blood vessels
Constriction

Liver
Stimulates bile release

Digestive system
Stimulates activity of
stomach and intestine

Uterus
Relaxation

Urinary system
Increases urinary output



Sympathetic

Pupil
Dilation

Heart
Increases heartbeat

Airways
Dilation

Sweat glands
Stimulates secretion

Liver
Stimulates release of
glucose

Digestive system
Decreases activity

Adrenal glands
Stimulates the production
of adrenaline

Uterus
Vaginal contraction

Urinary system
Relaxation

POST ASSESSMENT

Q1. Sympathetic outflow arises from:

- a) Thoracolumbar region
- b) Craniosacral region
- c) Lumbar region
- d) Cervical region

Q2. In the ANS, postganglionic neurons are located:

- a) In the brain
- b) In the spinal cord
- c) In autonomic ganglia
- d) In skeletal muscles

CONT..

Q3. All preganglionic neurons of the ANS release:

- a) Dopamine
- b) Acetylcholine
- c) Norepinephrine
- d) Serotonin

Q4. Which receptor type is primarily found at parasympathetic target organs:

- a) Adrenergic
- b) Nicotinic
- c) Muscarinic
- d) Dopaminergic